

Unit-1: Introduction to Online Social Networks

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PART 01

Introduction to Online Social Networks

What are Online Social Networks?

01

Definition and Key Features

OSNs are platforms where users create profiles, connect with others, and share content; they facilitate social interactions online, mirroring real-world relationships and communities.

02

Examples of Popular OSNs

Facebook, Twitter, Instagram, LinkedIn, and TikTok are examples; each cater to different social needs, from broad networking to professional connections and sharing creative content.

03

Evolution of Social Networks

Early platforms focused on basic connectivity; modern OSNs incorporate features like live video, e-commerce, and augmented reality, enhancing user engagement and platform capabilities.

The Rise of Social Media



Impact on Communication

OSNs have revolutionized communication by enabling instant global connections and democratizing content creation, reshaping how people interact and share information.



Influence on Culture and Society

Social networks shape cultural trends, influence political discourse, and affect social movements, with both positive and negative consequences observed globally.



Commercial Significance

Businesses leverage OSNs for marketing, advertising, and customer engagement; creating opportunities for brand visibility and direct consumer interaction.

PART 02

Data Collection from Social Networks

Methods for Data Extraction



01

Social Media APIs

APIs provided by OSNs allow developers to extract data such as user profiles, posts, and network connections; enabling research and application development.

02

Web Scraping Techniques

Web scraping involves extracting data directly from website HTML; this method is used when API access is limited, but it requires careful ethical and legal considerations.

03

Ethical Considerations in Data Collection

Collecting data must respect user privacy, adhere to terms of service, and comply with data protection regulations like GDPR, ensuring responsible use of personal information.

Types of Data Collected



User Profile Information

Username, demographics, interests, and social connections are commonly collected; providing insights into user behaviors and network structures for research and analysis.



Content Data

Posts, comments, likes, and shares represent the content shared by users; their analysis can reveal trends, sentiments, and influence within the network.



Network Structure

Relationships between users, such as friendships and followers, define structures; network analysis identifies influential actors and community clusters.

Tools for Data Analysis



01 Network Analysis Software

Gephi and UCINET are used to visualize and analyze network structures; they identify key nodes, communities, and influential members.



02 Sentiment Analysis Tools

Tools like VADER and TextBlob analyze text data to determine the sentiment expressed; they can track public opinion and emotional responses to events or topics.

PART 03

Challenges in Online Social Networks

Privacy Concerns

01

Data Breaches and Leaks

Data breaches expose user information, undermining trust in OSNs; examples include the Cambridge Analytic a scandal, where personal data was misused.

02

Privacy Settings and Management

Users struggle to manage their privacy settings effectively; most haven't configured advanced privacy setting; leading to unintentional exposure of personal information.

03

Data Security Measures

OSNs invest in data security to protect user information; they are still vulnerable to sophisticated cyberattacks that can compromise privacy.

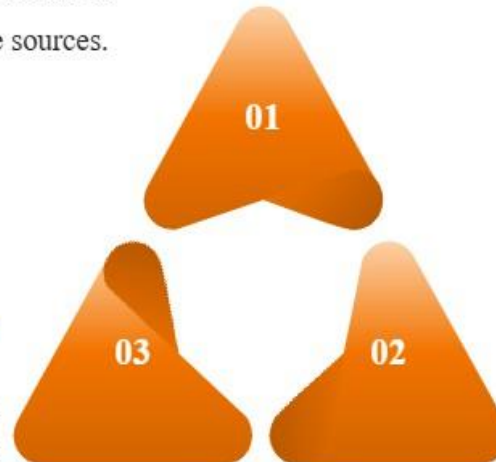
Misinformation and Fake News

Spread of False Information

OSNs facilitate the rapid spread of misinformation and fake news; impacting public opinion and trust in reliable sources.

Detection and Mitigation Strategies

Fact-checking initiatives, AI-based detection tools, and content moderation are used to combat misinformation; although these are difficult to scale.



Algorithms and Echo Chambers

Algorithms can create echo chambers, reinforcing existing beliefs; users are primarily exposed to information confirming their viewpoints.



PART 04

Opportunities in Online Social Networks



Enhanced Connectivity

Global Communication

OSNs connect people across geographic boundaries, facilitating global conversations and collaboration; this enhanced connectivity drives innovation.

01

Community Building

OSNs enable interest-based communities, fostering support networks and shared experiences; these communities promote identity formation.

02

Social Activism and Movements

OSNs support social activism by organizing events, amplifying voices, and mobilizing support for causes; this fosters collective action.

03

Business and Marketing



Targeted Advertising

OSNs enable targeted advertising, improving ad relevance and ROI for businesses; data analysis allows precision based ad campaigns.



Customer Engagement

Businesses use OSNs to engage with customers, providing support and building relationships; which can improve customer loyalty and brand image.



E-commerce and Social Shopping

OSNs integrate e-commerce features, enabling direct sales and social shopping experiences; creating new revenue streams for businesses.

Research and Education

01

Data Analysis for Research

OSN data is analyzed to study human behavior, social trends, and network dynamics; which enriches scientific discovery.

02

Educational Platforms

OSNs are used for educational purposes, facilitating online learning and knowledge sharing; it promotes collaborative learning environments.

PART 05

Pitfalls of Online Social Networks

Addiction and Mental Health



Excessive Usage

Excessive OSN usage can lead to addiction, affecting mental health; this is caused by dopamine feedback loops.



Cyberbullying

OSNs can be platforms for cyberbullying, causing emotional distress; the relative anonymity emboldens toxic interactions.



Social Comparison

Constant exposure to idealized images can lead to social comparison, affecting self-esteem; this affects self-image and body positivity.

Privacy Violation

Stalking

Platforms enable stalking, causing harm and invasion of space; users are often unaware of risk factors involved.



Identity Theft

OSNs are used to steal identities, leading to financial losses; creating user ID's and passwords that are vulnerable to phishing attacks.



Reduced Productivity

Procrastination

OSNs contribute to procrastination, impacting productivity, especially in schools and at the workplace; distraction degrades performance on tasks.



PART 06

Social Media APIs for Data Extraction

Introduction to APIs

01

What is an API?

An API (Application Programming Interface) enables software applications to interact and exchange data; facilitating efficient information exchange.

02

Role of APIs in Data Extraction

APIs provide structured access to OSN data, enabling developers to collect and analyze information efficiently; streamlining structured data acquisition.

03

Types of Social Media APIs

REST APIs, GraphQL APIs, and Streaming APIs offer different functionalities for data extraction; each method affects speed and the data types accessible.

Popular Social Media APIs



Twitter API

The Twitter API allows developers to access tweets, user profiles, and trends.



Facebook Graph API

The Facebook Graph API provides access to user profiles, posts, and connections, enabling a wide range of applications.



Instagram API

The Instagram API facilitates access to photos, videos, and user data; which supports analysis of visual culture.

Using APIs for Research

01

Case Studies

Researchers use APIs to study topics like public opinion, social movements, and information diffusion; it supports data driven insight generation.

02

Best Practices

Adhering to rate limits, respecting terms of service, and ensuring data privacy are crucial when working with APIs, allowing research within ethical guidelines.

PART 07

Conclusion

Summary of Key Points



Recap of OSN Definition and Evolution

OSNs have revolutionized communication and social interaction, evolving rapidly over time; this evolution has created challenges and opportunities.

Overview of Data Collection, Challenges, and Opportunities

Understanding data collection methods, addressing challenges like privacy, and leveraging opportunities are crucial in the OSN landscape; each affects social and economic impacts.



Future Trends



Emerging Technologies

Augmented reality, blockchain, and AI are transforming OSNs, creating immersive and personalized experiences; these evolving technologies promise innovation.



Evolving Social Media Landscape

OSNs will continue to evolve, adapting to user needs and technological advancements; influencing the direction of online interaction.

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